NETWORKING

Networking In a Silo is Not Enough - Building End-to-End Automation for AI Factories

The IREN Use case



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OPEN STARTUP™

About IREN

VERTICALLY INTEGRATED AI CLOUD

IREN develops, builds and operates its own data centers In-house engineering.

Proven delivery track record, consistently meeting targets

Years of de-risking supply chains, sites, people and processes is a competitive advantage

>4.5GW Secured Power

810MW Operating Data Centers

$2.3bn Contracted AI Revenue

Customer validation from leading global counterparts

2,000+ Founder-led Team

Vertically Integrated AI Cloud

Vertically Integrated AI Cloud	**Land** **Grid-Connected Power**
Internal Construction & Procurement Teams	**Data Centers** **Compute**
Internal Technology & Operations Teams	**Technology Stack**

Current IREN Deployments

VERTICALLY INTEGRATED AI CLOUD

New 1.6GW data center campus secured in Oklahoma

Expanded >4.5GW secured power portfolio broadens customer universe

Enables deeper, parallel commercial discussions anchored around a growing portfolio of firm capacity

Active DCs: Power/(CPUs/GPUs)/Eth vs IB

DCs in construction: Power/(GPUs)/Eth vs IB

Technology evolution – move to B300/liquid cooled



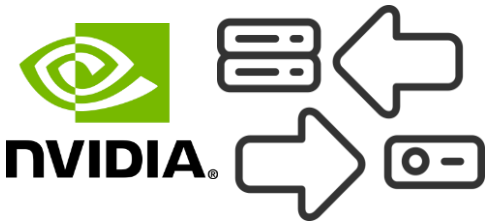
+ multi-GW development portfolio

Oklahoma site: Attractive for large-scale AI infrastructure

- **Grid studies complete**
- **2,000 acres secured**
- **Power scheduled to ramp from 2028**
- **<6ms latency to network hubs**

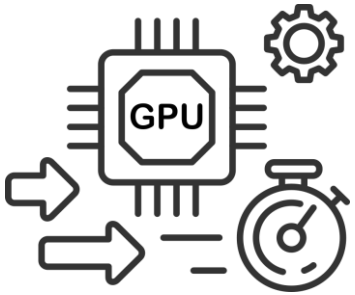
IREN

Current IREN Deployments



- NVIDIA RA deployment with ZERO deviations

- Achieves breakthrough performance levels.



- Bare metal to tokens ready network in under 2 hours

NVIDIA ⟷ BROADCOM

AMD

- Silicon diversity – NVIDIA, Broadcom, AMD & More

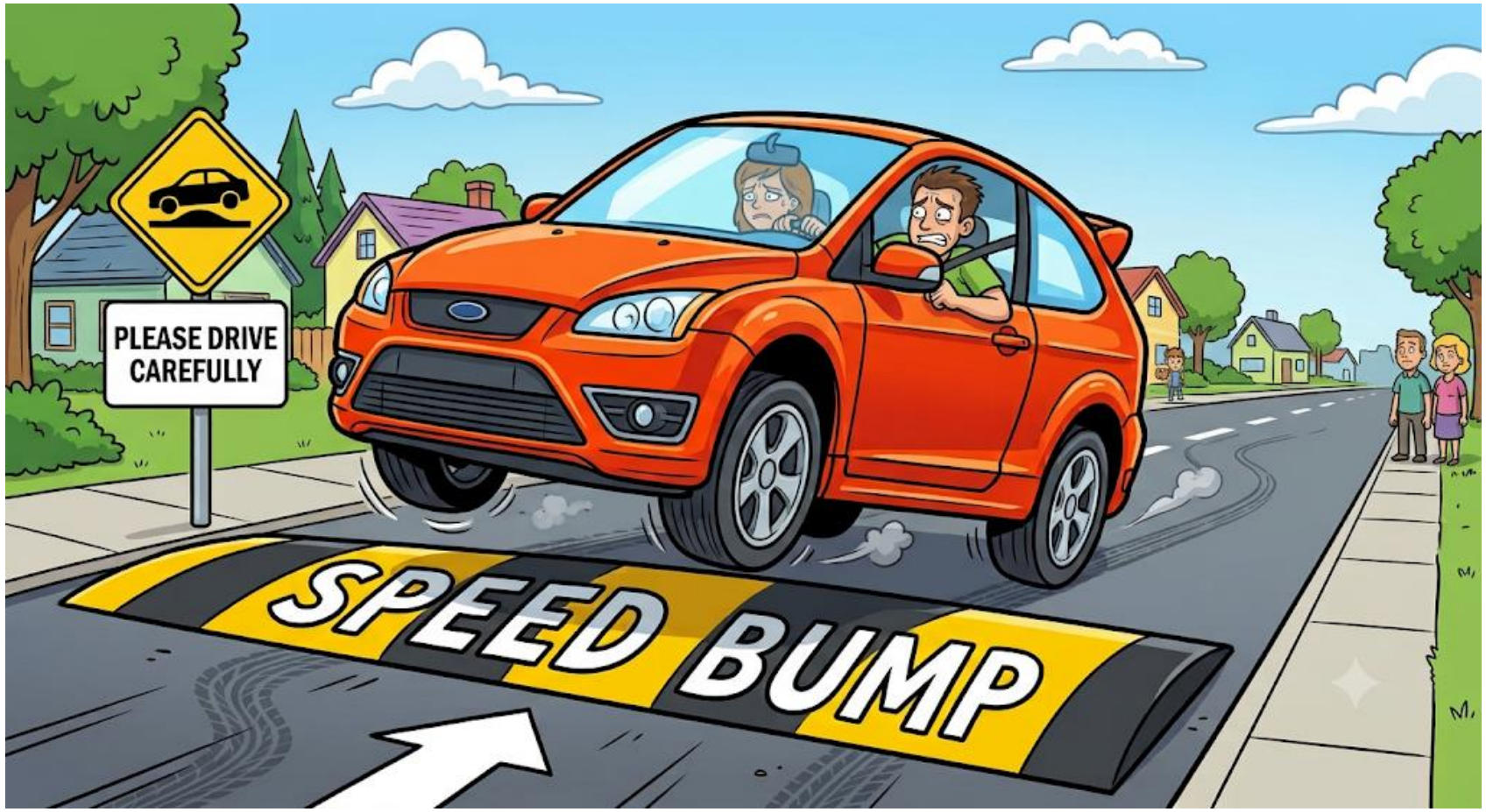


IREN deployment scenario:

- Migration from Cisco to SONiC with Broadcom ASIC
- Infiniband with Spectrum-X Ethernet focus
- Dell SONiC on Spectrum-X
- AMD deployments with Broadcom ASIC and SONiC

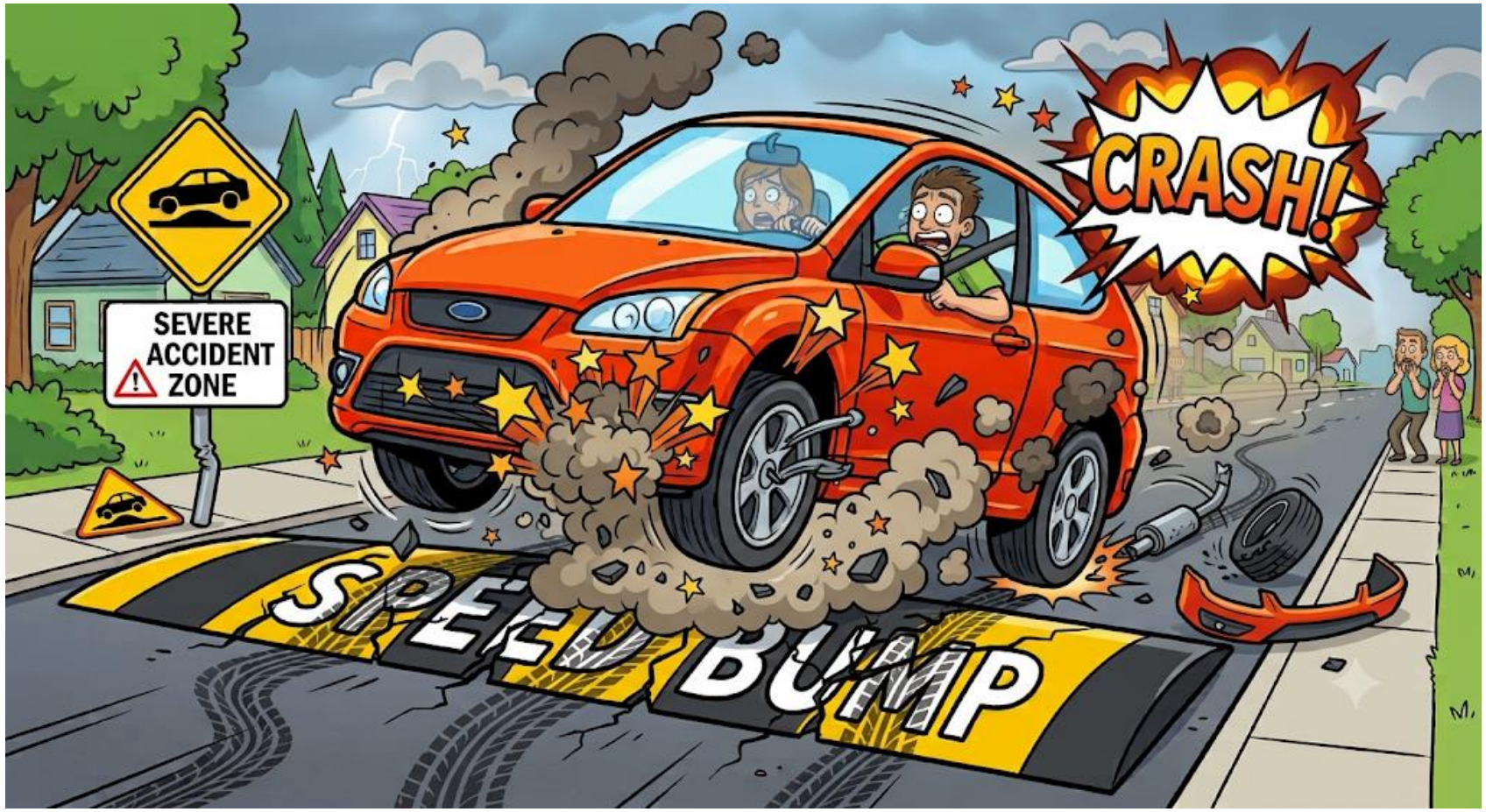
Networking Infrastructure Challenges

Networking issues are like speedbumps… In traditional workloads it is like hitting the speedbump at 30 mph… Not optimal but **survivable**



Networking Infrastructure Challenges

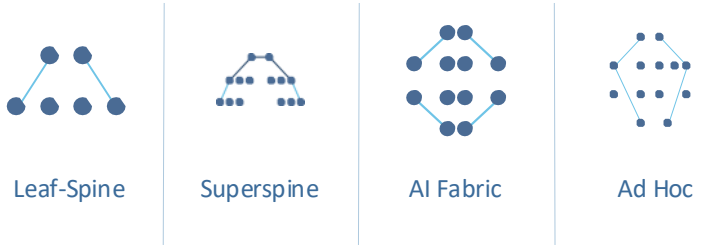
In the age of AI it is like hitting the same bump at 100 mph – ***catastrophic***



The Intelligent Infrastructure Management Platform

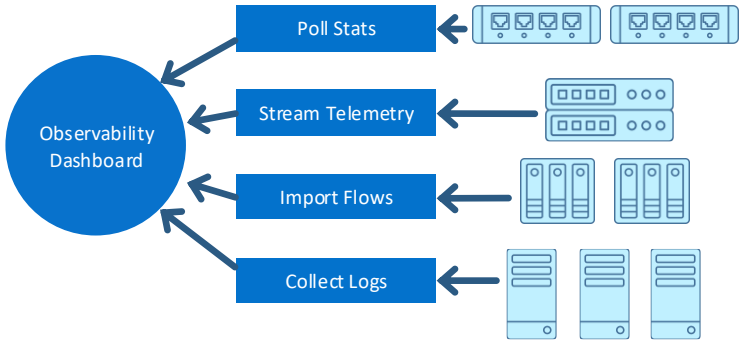
Unifies Automation, Observability, and Agentic AI to take customers from Day 0 deployment through ongoing operations

Automation



- Intent in: RA or design JSON
- Fabric out: NS, EW, OOB
- Live topology validated against intended design

Observability



- Streaming telemetry per port, link, and device
- Alarms on CRC / FEC and other fabric errors vs baseline
- One view across the fabric

Agentic AI



- Runs on-prem; nothing leaves the site
- MCP: JSON + discovered fabric in, missing/wrong cables out
- Human approves before anything is pushed

Lesson 1: Intended Design vs. Live Fabric

What We Had:

- EW bring-up: AMD GPUs on Micas HW /Broadcom SONiC, no reference architecture
- NS with BRCM SONiC+Celestica HW
- Intent was a spreadsheet of spine, leaf, and server ports
- BE to discover the live fabric

What We Did:

- Spreadsheet + discovered topology into MCP (link pattern recognition)
- Agentic AI flagged missing and mismatched cables
- IREN fixed the physical plant from that view



- **Without a coded RA, BE could not auto-alert on every mis-cable. The wide view still beat a per-domain hunt.**
- **Next deployments load a real RA so BE automatically alerts when the physical fabric drifts from intent.**

Lesson 2: Optics and Lossless are Day 0

CRC / FEC Errors

BE alarms when link errors break the baseline. Dirty or damaged fiber showed up here first.

Fiber

CRC and FEC spikes pointed at specific links. Those strands were inspected, cleaned, or replaced before training traffic ran on them.

RoCE Tuning

BE pushed one RoCE profile onto the AMD GPUs and the SONiC switches: queues, ECN, PFC. East-west stays lossless. GPU-to-GPU traffic stays off the slow path.



Lesson 3 — From JSON to Reference Architecture

First Deployment

- Spreadsheet uploaded to BE to build the Broadcom/SONiC underlay
- No RA, so no automatic drift alert

Next Deployments

- Childress / MacKenzie: NVIDIA Spectrum-X, NCP RA 2.1.2
- RA is intent BE can enforce
- Physical fabric veers from design, BE alerts



A JSON lets BE provision the underlay but an RA lets BE compare the live fabric to the design and auto-alert on drift.

Market Evolution/Revolution

Humans Are Not Enough - Unified System Management Is Now Strategic Imperative

AI Density – Infrastructure Has Outgrown Its Management Model

From Tens of Node @ 5-10 kW to Thousands of Nodes @80-140 kW

- AI scale environments demand real-time topology intelligence, dynamic power and cooling – period polling and siloed tools cannot keep pace
- Management software is no longer optional plumbing – it is mission-critical for every deployment. Customers struggling to operate these large scale and complex environments

The User/Buyer – The Priorities Have Evolved

From IT Admin to Proactive Infrastructure Management

- Buying behavior heavily determined by "Time-to-First Outcome/Token", Trust of Technology, Validated Solutions and operational simplicity
- Hi-velocity operators (Neoclouds, Fintech, large AI first enterprises, etc.), demand unified management tool across all infra, MCP, real-time telemetry, agentic interface and AI remediation

Competitive Landscape – Software Becoming the Primary Battleground

Innovation | Agility | Profitability | Adaptability

- NVIDIA and AMD are aggressively moving up the software stack into system and control solutions
- OEMs like Dell, HPE, SuperMicro and others want to differentiate and protect their offerings and not being commoditized
- Software solutions offer a steady stream of high-margin revenues/cashflows

The Intelligent Infrastructure Management Platform

Unifies Automation, Observability, and Agentic AI to take customers from Day 0 deployment through ongoing operations

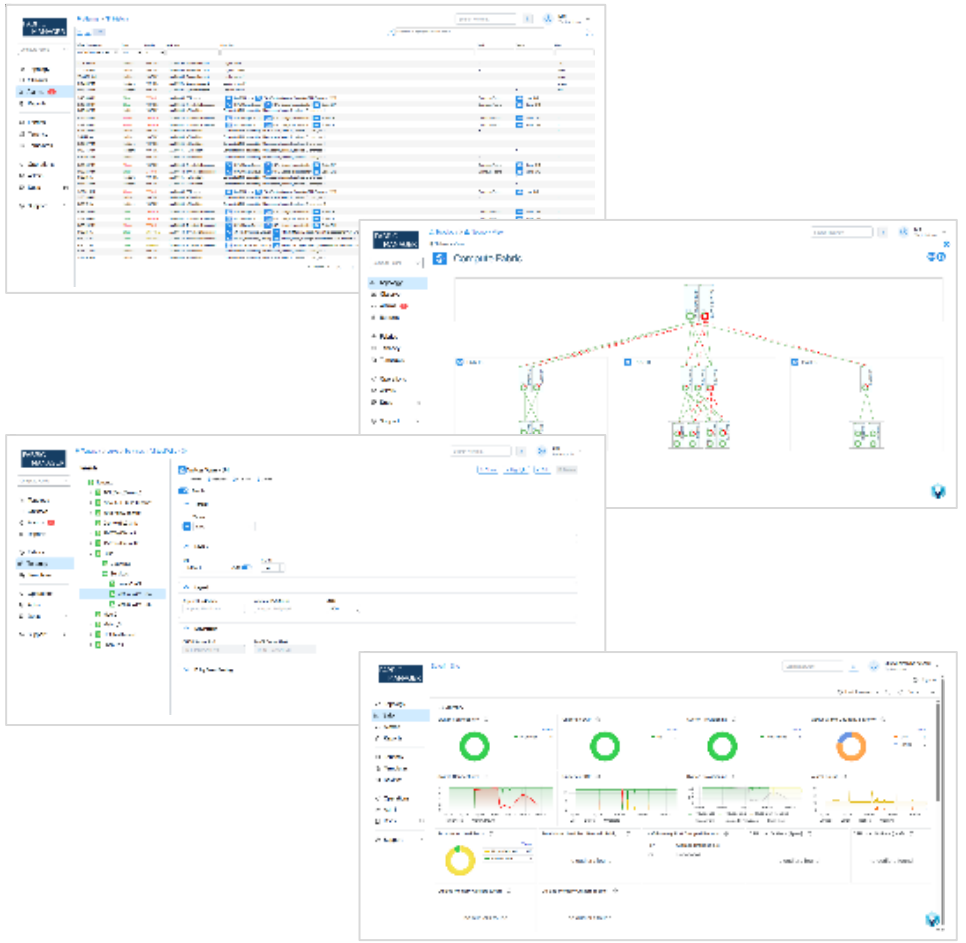
From Reactive Monitoring → Proactive → Predictive → Prescriptive → Autonomous Operations

Core Architecture

- **On-premise** with local LLM for data privacy and security
- Data Fabric – open and standards based
- AI Fabric: DSPy framework, MLflow tracking
- Graph database for situational awareness
- Telemetry streaming and time-series DB
- E2E Infra with MCP Integration between parts
- Close collaborations between silicon, infra and model vendors
- Agentic AI conversational AI assistant – ***ENGLISH IS THE NEW CODING LANGUAGE***

Customer Deployment

1 Data Foundation	2 ML Analytics	3 Prescriptive RCA	4 Autonomous Actions
Unified telemetry collection & standardization	Anomaly detection & predictive alerts	AI-driven root cause & remediation guidance	Self-healing with HITM approval



95% MTTR Reduction	100% Data Privacy	**Three Pillars** Metrics, Logs &Traces	**AI-First** Intelligent Automation

Scaling into the future



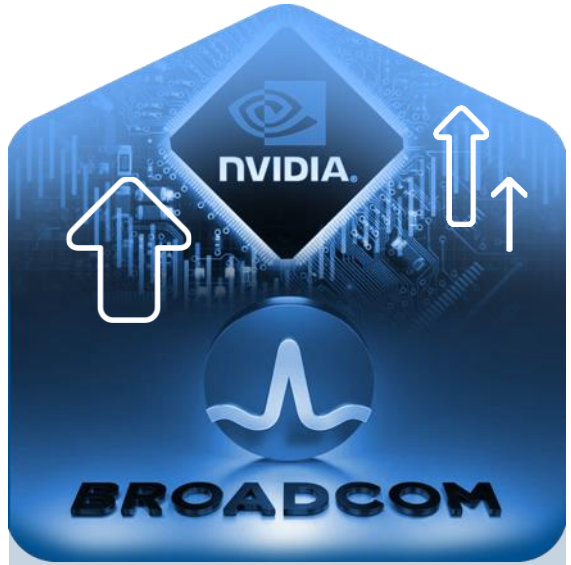
Automate everything with planning for immediate growth



Couple the network to the software stack (zero operational workflow gaps)



Go deeper on OCP based HW and related projects



Standardize network on two silicon – Broadcom and Spectrum-X

Call to Action:

Share Your Fabric 'Lessons Learned' with the OCP Networking Project

IREN just walked you through what worked and what it took to get there.

Every company running AI fabric has its own version of this story. Bring yours to the group that's working on these problems.

1. **Subscribe to the mailing list**
   OCP-Networking@OCP-All.groups.io

2. **Join the monthly call**
   2nd Monday, 10–11am PT — open to the public

3. **Request time on the agenda**
   Contact the project leads listed on the OCP Networking wiki page



Open Discussion



Thank You!

